ASSIGNMENT 1 • UNIVERSITY PRESENTATION

Agentic AI Case Study: Klarna AI Copilot

CASE STUDY

COMPANY
Klarna

INDUSTRY
Fintech / Financial Services

USE CASE
AI Copilot for internal workflows

blog.langchain.dev/customers-klarna/

PRESENTED BY

NAME
Munirah Alwadani

COMPANY
[NASSAQ]

Klarna Bank AB

Global Fintech Leader

Klarna is a global payments and shopping service that provides smarter, more flexible financial solutions to consumers and merchants. Known for disrupting the credit industry with "Buy Now, Pay Later" (BNPL).

85M+
ACTIVE USERS

2.5M
DAILY TRANSACTIONS

45+
MARKETS

500k+
MERCHANTS

Core Services

- Payments & Checkout Solutions
- Buy Now Pay Later (BNPL) Financing
- Merchant Growth Tools & Marketing
- Consumer Banking & Cards

Why AI is Critical

With millions of daily transactions, manual workflows for support, risk, and compliance are impossible to scale linearly.

Goal: Unify complex financial workflows & reduce manual triage

Notes: Klarna is a major fintech player handling massive volume. The scale (2.5M tx/day) creates operational bottlenecks that traditional software alone can't solve efficiently—hence the need for Agentic AI.

The Problem: Operational Complexity

Fragmented Infrastructure

Employees must navigate a complex web of disconnected internal tools to resolve even simple queries. Data is siloed, requiring manual retrieval.

Payment Databases
Transaction ledgers & histories

Risk Engines
Fraud detection & scoring

Compliance Rules
Regulatory policy checks

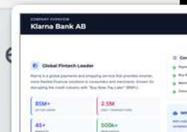
Customer CRM
Profiles & support tickets

Workflow Bottlenecks

1. **Manual Escalation Overload**
   Multi-departmental issues require passing tickets between teams, causing massive delays.
2. **Context Switching Fatigue**
   Agents waste time logging in and out of different dashboards to gather a complete picture.
3. **Scalability Limits**
   Linear hiring cannot keep up with exponential transaction growth (2.5M daily).

"Manual workflows act as a brake on growth. Agents are stuck acting as 'human APIs' copy-pasting data between systems."

Notes: The core problem isn't just volume; it's complexity. Data lives in silos (Payment DBs, Risk Engines). Human agents act as "glue" between these systems, which is slow, e
prone, and doesn't scale.



THE AI SOLUTION

Klarna AI Copilot

Tech Stack

LangGraph
Orchestrates complex, stateful multi-agent workflows and routing.

LangSmith
Provides testing, evaluation, and monitoring for reliability.

Core Capabilities

Natural Language Interface
Allows employees to interact with complex internal systems using plain English.

Autonomous Task Execution
Handles payments, refunds, and escalations by retrieving data and triggering actions.

Policy Adherence
Strictly follows compliance rules and guardrails during execution.

Notes: The solution is an AI Copilot that doesn't just "chat" but acts. It uses LangGraph to plan multi-step workflows (like risk analysis) and connects natural language requests to deterministic internal syst

ARCHITECTURE

System Architecture: Agent vs. Infrastructure

AI / Agent Layer

THE "BRAIN" & ORCHESTRATOR

LLM Reasoning Engine

Responsible for planning, multi-step reasoning, and selecting which tools to use.

LangGraph Controller

Manages state, loops, and multi-agent coordination.

Context-Aware Prompts

Dynamically tailors instructions based on scenario policies.

LangSmith Evaluation

Monitors performance, traces errors, and validates outputs.

Calls Tools / APIs

Returns Data / Status

Traditional Software

THE "TOOLS" & INFRASTRUCTURE

Internal APIs

Deterministic interfaces for payments, refunds, and user profiles.

Databases

Transactional systems of record (SQL/NoSQL) holding customer data.

Risk & Compliance Engines

Hard-coded rule engines for fraud detection and legal checks.

Reporting Systems

Email gateways, ticket creation systems, and BI dashboards.

Notes: The "Agent" isn't replacing the software; it's orchestrating it. The Left side (Blue) thinks and plans. The Right side (Grey) executes deterministic actions. The arrows show the continuous loop of Action → Observation. This separation is crucial for stability in enterprise environments.

AGENT BEHAVIOR

What is "Agentic" in this System?

Dynamic Tool Selection

The LLM acts as a router, autonomously deciding which internal API or database to query based on the user's intent, rather than following a hard-coded path.

Multi-Step Reasoning

The agent breaks down complex requests (e.g., "refund and notify") into a logical sequence of operations, executing them step-by-step.

Self-Correction

If an API call fails or returns ambiguous data, the agent can analyze the error and retry with different parameters without human intervention.

Multi-Agent Routing

Using LangGraph, the system routes tasks to specialized agents (e.g., a "Refund Specialist" vs. "Compliance Officer") to handle distinct domains.

State Management

The system maintains "state" across the conversation, remembering previous outputs to inform future actions within the same workflow.

Context-Aware Planning

The agent dynamically adjusts its plan based on real-time data retrieved from tools, handling edge cases that weren't explicitly pre-programmed.

CORE DEFINITION

> *"This qualifies as **Agentic AI** because the system autonomously reasons about how to achieve a goal using available tools, rather than just following a pre-defined script."*

NOTES

The key differentiator is autonomy in the "how". Traditional software executes a known path. The Agentic Copilot figures out the path (Tool Selection, Reasoning) and adapts if th
go wrong (Self-Correction). It doesn't just retrieve information; it actively plans and executes multi-step workflows.

What is NOT Agentic?

Traditional Components

Databases & Ledgers

Stores of truth for payments, transaction history, and customer profiles. Passive data repositories.

Compliance Rule Engines

Hard-coded logic (if-this-then-that) for regulatory checks. Deterministic and unbending.

Internal APIs

REST/GraphQL endpoints that execute specific actions (e.g., "ProcessRefund").

Reporting Tools

Dashboards and BI tools that display data but do not make decisions or take actions.

Why NOT Agentic?

Deterministic Behavior

Input A always equals Output B. No reasoning, planning, or ambiguity is involved.

Access-Controlled & Audited

Rigid permissions and audit trails. These systems do not "decide" to access data; they are accessed.

Spec-Defined Interfaces

Operate via strict API contracts (OpenAPI/Swagger). They cannot "figure out" how to work if inputs are wrong.

The "Tool" Analogy

These systems are the **tools** in the toolbox. The AI Agent is the **worker** who decides which tool to pick up and how to use it to solve a problem.

Notes:

It is crucial to distinguish the "brain" from the "body". The databases, rule engines, and APIs represent the robust, deterministic infrastructure of the bank. They are NOT agentic because they lack autonomy, planning capabilities, and reasoning; they simply execute specific commands reliably when called upon. The agentic system sits on top of this laye orchestrating these tools to achieve complex goals.

Benefits to Klarna

> *By deploying an Agentic AI system that orchestrates internal tools and workflows, Klarna achieved massive efficiency gains that linear automation could not match.*

80%

FASTER RESOLUTIONS

Reduction in average time to resolve customer queries and internal escalations compared to previous workflows.

70%

AUTOMATION RATE

Of repetitive support tasks are now handled entirely by the AI agent without human intervention.

700

FTE EQUIVALENT

The AI agent performs the work output equivalent to 700 full-time human service agents.

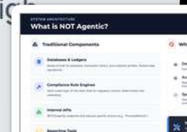
Employee Productivity

Reduced context switching by allowing employees to interact with complex backend systems via natural language instead of multiple dashboards.

Accuracy & Consistency

Systematic evaluations via LangSmith ensure consistent adherence to compliance policies across all interactions.

NOTES: The business impact is quantifiable and massive. Key metrics include 80% faster resolution times and work output equivalent to 700 full-time staff members. This proves that Agentic AI isn't just a novelty; it's a core productivity driver that fundamentally changes unit economics. The 70% automation rate means human agents can focus entirely on high-value, complex cases that require empathy and judgment, while the AI handles the routine volume.



CRITICAL ANALYSIS

Risks and Limitations

Hallucinations & Policy Drift

LLMs can confidently generate incorrect financial advice or misinterpret complex internal policies.

MITIGATION STRATEGY

Systematic evaluations & testing (LangSmith)

Incorrect Tool Invocation

Agents may call the wrong API endpoint, pass incorrect parameters, or trigger unintended actions in backend systems.

MITIGATION STRATEGY

Strict schema validation & read-only modes

Security & Data Privacy

Risk of exposing PII (Personally Identifiable Information) to the LLM provider or retrieving unauthorized data from internal DBs.

MITIGATION STRATEGY

PII masking & RBAC (Role-Based Access Control)

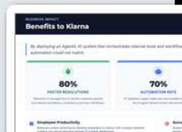
Need for Human Oversight

Autonomous agents can compound errors if left unchecked. Complex financial decisions require accountability.

MITIGATION STRATEGY

Human-in-the-loop (HITL) approval steps

Notes: While powerful, Agentic AI introduces non-deterministic behavior into financial systems. Mitigations like LangGraph for control flow and LangSmith for evaluation are critical for safe deployment. It is essential to understand that agents are probabilistic engines, not deterministic ones like traditional code, so guardrails are non-negotiable in fintech.



Key Takeaways

1 Agentic AI = Reasoning + Tools
It's not just a chatbot. The core value is the combination of an LLM's reasoning capabilities with the ability to execute actions via external tools.

2 Dependency on Traditional Infrastructure
Enterprise AI cannot function in a vacuum. It relies heavily on robust, deterministic traditional software (APIs, DBs) to actually perform work.

3 Orchestration over Replacement
Agents act as an orchestration layer on top of existing systems, rather than replacing the underlying transactional engines.

4 Separation of Concerns
Keeping the "reasoning" (AI) separate from the "execution" (APIs) improves safety, auditability, and system reliability.

5 Safety First
Adopting Agentic AI requires rigorous monitoring, guardrails (like LangSmith), and human-in-the-loop protocols to mitigate hallucination risks.

> *"The future of enterprise software isn't just about better interfaces, but about **autonomous agents** that can reason, plan, and execute complex workflows on our behalf."*

NOTES

In summary, Klarna's success serves as a pivotal case study demonstrating that Agentic AI acts as a powerful orchestration layer rather than a standalone replacement. It doesn't replace your database or core APIs; instead, it makes them smarter, more accessible, and capable of handling complex, multi-step workflows autonomously. The key takeaway for future implementations is the clear separation between the reasoning engine (the agent) and the deterministic execution tools (the infrastructure).